Terraform AWS S3 Security Lab with Checkov

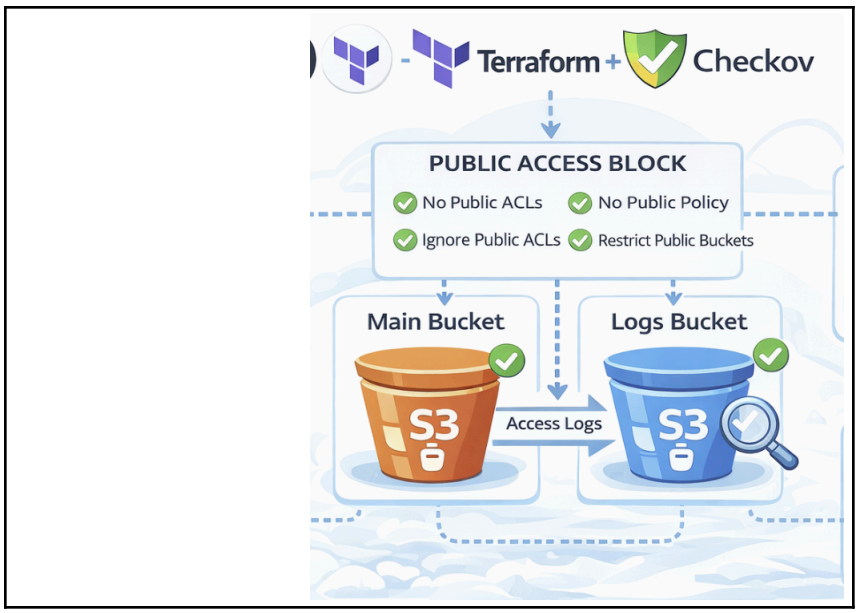

Table of Contents

1 Objectif du lab..... 2

2 Requis :..... 2

2.1 Infrastructure as Code..... 2

2.2 Cloud Provider Access..... 3

2.3 Security Scanning Tool..... 3

2.4 Runtime Environment..... 3

3 Structure du projet..... 4

4 provider.tf..... 5

5. variables.tf..... 5

6. main.tf (VERSION VOLONTAIREMENT NON SÉCURISÉE)..... 5

7. outputs.tf..... 6

8. environments/dev/terraform.tfvars..... 6

9. .checkov.yaml..... 6

10. README.md....6
11. WORKFLOW COMPLET....7
🟡 Étape 1 — Initialisation....7
🟡 Étape 2 — Plan....7
🟡 Étape 3 — Scan sécurité....8
📊 Résultat attendu....10
🟡 Étape 4 — Correction (DevSecOps loop)....11
4.1 Correction – première étape....11
4.2 CKV_AWS_18....11
4.3 CKV2_AWS_61 — Lifecycle....12
4.4 autres corrections optionnelles....12
4.5 Enlever des exceptions....12
4.6 tests finaux :....13
🟡 Étape 5 — Apply....15
Annexe 1 – version sécurisée....16
Annexe 2 – Version finale....16

1 Objectif du lab

Construire un workflow local :

- Terraform (infra simple mais réaliste)
- Scan sécurité avec Checkov
- Boucle : écrire → plan → scanner → corriger → appliquer

2 Requis :

2.1 Infrastructure as Code

- **Terraform** ≥ 1.5
  Used to provision AWS resources (S3 bucket)

👉 Verify installation:

```
terraform -version
```

2.2 Cloud Provider Access

- **AWS CLI** configured with valid credentials

👉 Configure access:

```
aws configure
```

👉 This lab requires permissions to:

- create S3 buckets
- manage bucket configurations (encryption, versioning, policies)

2.3 Security Scanning Tool

- **Checkov**

py -3.11 -m pip install checkov

👉 Verify:

```
checkov –version
```

```
C:\Users>checkov --version
File association not found for extension .py
3.2.513
```

2.4 Runtime Environment

- **Python 3.8+** (required for Checkov)
- **Ici dasn ce lab on utilise la version 3.11.9**

👉 Verify:

```
python –version ou bien py -0
```

Dans notre OS , on a deux versions de Python installés.

On utilisera la version 3.11 (la version 3.13 présente des erreurs avec checkov)

```
 -V:3.13 *        Python 3.13 (64-bit)
PS C:\terraform\terraform-checkov-lab> py -0
 -V:3.13 *        Python 3.13 (64-bit)
 -V:3.11          Python 3.11 (64-bit)
PS C:\terraform\terraform-checkov-lab>
```

:

---

3 Structure du projet

```
terraform-checkov-lab/
├── provider.tf
├── main.tf
├── variables.tf
├── outputs.tf
│
├── environments/
│   └── dev/
│       └── terraform.tfvars
│
├── .checkov.yaml
└── README.md
```

```
    Directory: C:\terraform\terraform-checkov-lab


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         3/27/2026   9:12 PM                .terraform
d-----         3/27/2026   8:51 PM                environments
-a----         3/27/2026   9:12 PM           1407 .terraform.lock.hcl
-a----         3/27/2026   8:56 PM             42 checkov.yaml
-a----         3/27/2026   8:50 PM            443 main.tf
-a----         3/27/2026   8:51 PM             60 outputs.tf
-a----         3/27/2026   8:46 PM            209 provider.tf
-a----         3/27/2026   8:48 PM            255 variabes.tf


PS C:\terraform\terraform-checkov-lab>
```

4 provider.tf

```
terraform {
  required_version = ">= 1.5.0"

  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

provider "aws" {
  region = var.aws_region
}
```

---

5. variables.tf

```
variable "aws_region" {
  description = "AWS region"
  type        = string
}

variable "bucket_name" {
  description = "S3 bucket name"
  type        = string
}

variable "tags" {
  description = "Common tags"
  type        = map(string)
}
```

---

6. main.tf (VERSION VOLONTAIREMENT NON SÉCURISÉE)

👉 Ici on veut que Checkov trouve des erreurs

```
resource "aws_s3_bucket" "this" {
  bucket = var.bucket_name
```

```
  tags = var.tags
}

# ❌ Bucket public (grosse fail sécurité)
resource "aws_s3_bucket_public_access_block" "this" {
  bucket = aws_s3_bucket.this.id

  block_public_acls       = false
  block_public_policy     = false
  ignore_public_acls      = false
  restrict_public_buckets = false
}

# ❌ Pas de versioning
# ❌ Pas de encryption
# ❌ Pas de logging
```

---

7. outputs.tf

```
output "bucket_name" {
  value = aws_s3_bucket.this.id
}
```

---

8. environments/dev/terraform.tfvars

```
aws_region  = "ca-central-1"
bucket_name = "henri-checkov-lab-123456"

tags = {
  Environment = "dev"
  Project     = "terraform-checkov-lab"
}
```

---

9. .checkov.yaml

```
framework:
  - terraform

output: cli
```

10. README.md

```
# Terraform + Checkov Lab (S3)

## Objective
Demonstrate a DevSecOps workflow using Terraform and Checkov locally.

## Steps
1. terraform init
2. terraform plan
3. checkov -d .
4. Fix issues
5. terraform apply
```

11. WORKFLOW COMPLET

🟡 Étape 1 — Initialisation

🟡 `terraform init`

👉 Télécharge provider AWS

```
PS C:\terraform\terraform-checkov-lab> terraform init
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

🟡 Étape 2 — Plan

```
terraform plan -var-file="environments/dev/terraform.tfvars"
```

👉 Vérifie ce qui va être créé

```
  # aws_s3_bucket_public_access_block.this will be created
  + resource "aws_s3_bucket_public_access_block" "this" {
      + block_public_acls       = false
      + block_public_policy     = false
      + bucket                  = (known after apply)
      + id                      = (known after apply)
      + ignore_public_acls      = false
      + restrict_public_buckets = false
    }

Plan: 2 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + bucket_name = (known after apply)
```

🟡 Étape 3 — Scan sécurité

```
(jai eu des problèmes d'encodage au niveau du format et ai dû utiliser la
variable d'environnement suivante positionnée, afin d'utiliser le codage UTF8

 $env:PYTHONUTF8=1

J'ai utilisé la version v3.11 de python, car la version v3.13 ne fonctionnait
pas avec checkov. Donc utiliser la commande py -3.11 - m, ci-dessous

 py -3.11 -m checkov.main -d .
```

👉 Lance Checkov

```
PS C:\terraform\terraform-checkov-lab>
PS C:\terraform\terraform-checkov-lab> $env:PYTHONUTF8=1
PS C:\terraform\terraform-checkov-lab> py -3.11 -m checkov.main -d .
```

Les résultats sont très verbeux (tient sur plusieurs pages d'écran)

```
PS C:\terraform\terraform-checkov-lab> $env:PYTHONUTF8=1
PS C:\terraform\terraform-checkov-lab> py -3.11 -m checkov.main -d .
[ kubernetes framework ]: 100%|                    |[1/1], Current File Scanned=checkov.yaml
[ terraform framework ]: 100%|                    |[4/4], Current File Scanned=variabes.tf
[ secrets framework ]: 100%|                    |[5/5], Current File Scanned=.\variabes.tf


       _               _
   ___| |__   ___  ___| | _______   __
  / __| '_ \ / _ \/ __| |/ / _ \ \ / /
 | (__| | | |  __/ (__|   < (_) \ V /
  \___|_| |_|\___|\___|_|\_\___/ \_/

By Prisma Cloud | version: 3.2.513

terraform scan results:

Passed checks: 5, Failed checks: 11, Skipped checks: 0

Check: CKV_AWS_93: "Ensure S3 bucket policy does not lockout all but root user. (Prevent lockouts needing root account f
ixes)"
        PASSED for resource: aws_s3_bucket.this
        File: \main.tf:1-5
        Guide: https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/s3-policies/bc-aws-s3-24
Check: CKV_AWS_41: "Ensure no hard coded AWS access key and secret key exists in provider"
        PASSED for resource: aws.default
        File: \provider.tf:12-14
        Guide: https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/secrets-policies/bc-aws-s
ecrets-5
Check: CKV_AWS_19: "Ensure all data stored in the S3 bucket is securely encrypted at rest"
        PASSED for resource: aws_s3_bucket.this
        File: \main.tf:1-5
```

On peut filtrer avec la commande :

py -3.11 -m checkov.main -d . –quiet

ou bien :

py -3.11 -m checkov.main -d . –compact

je ne montre qu’une partie des résultats dans la figure ci-dessous

```
PS C:\terraform\terraform-checkov-lab> py -3.11 -m checkov.main -d . --compact
[ kubernetes framework ]: 100%|                    |[1/1], Current File Scanned=checkov.yaml
[ terraform framework ]: 100%|                    |[4/4], Current File Scanned=variabes.tf
[ secrets framework ]: 100%|                    |[5/5], Current File Scanned=.\variabes.tf


       _                   _
   ___| |__   ___  ___| | _______   __
  / __| '_ \ / _ \/ __| |/ / _ \ \ / /
 | (__| | | |  __/ (__|   < (_) \ V /
  \___|_| |_|\___|\___|_|\_\___/ \_/

By Prisma Cloud | version: 3.2.513

terraform scan results:

Passed checks: 5, Failed checks: 11, Skipped checks: 0

Check: CKV_AWS_93: "Ensure S3 bucket policy does not lockout all but root user. (Prevent lockouts needing root account
ixes)"
        PASSED for resource: aws_s3_bucket.this
        File: \main.tf:1-5
        Guide: https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/s3-policies/bc-aws-s3-24
Check: CKV_AWS_41: "Ensure no hard coded AWS access key and secret key exists in provider"
        PASSED for resource: aws.default
        File: \provider.tf:12-14
        Guide: https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/secrets-policies/bc-aws-
ecrets-5
Check: CKV_AWS_19: "Ensure all data stored in the S3 bucket is securely encrypted at rest"
        PASSED for resource: aws_s3_bucket.this
        File: \main.tf:1-5
        Guide: https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/s3-policies/s3-14-data-e
crypted-at-rest
Check: CKV_AWS_57: "S3 Bucket has an ACL defined which allows public WRITE access."
        PASSED for resource: aws_s3_bucket.this
```

```
Check: CKV_AWS_20: "S3 Bucket has an ACL defined which allows public READ access."
        PASSED for resource: aws_s3_bucket.this
        File: \main.tf:1-5
        Guide: https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/s3-policies/s3-1-acl-read
-permissions-everyone
Check: CKV_AWS_53: "Ensure S3 bucket has block public ACLS enabled"
        FAILED for resource: aws_s3_bucket_public_access_block.this
        File: \main.tf:8-15
        Guide: https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/s3-policies/bc-aws-s3-19
Check: CKV_AWS_54: "Ensure S3 bucket has block public policy enabled"
        FAILED for resource: aws_s3_bucket_public_access_block.this
        File: \main.tf:8-15
        Guide: https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/s3-policies/bc-aws-s3-20
Check: CKV_AWS_55: "Ensure S3 bucket has ignore public ACLs enabled"
        FAILED for resource: aws_s3_bucket_public_access_block.this
        File: \main.tf:8-15
        Guide: https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/s3-policies/bc-aws-s3-21
Check: CKV_AWS_56: "Ensure S3 bucket has 'restrict_public_buckets' enabled"
        FAILED for resource: aws_s3_bucket_public_access_block.this
        File: \main.tf:8-15
        Guide: https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/s3-policies/bc-aws-s3-22
Check: CKV2_AWS_62: "Ensure S3 buckets should have event notifications enabled"
        FAILED for resource: aws_s3_bucket.this
        File: \main.tf:1-5
        Guide: https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/aws-logging-policies/bc-a
ws-2-62
Check: CKV_AWS_145: "Ensure that S3 buckets are encrypted with KMS by default"
        FAILED for resource: aws_s3_bucket.this
        File: \main.tf:1-5
        Guide: https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/aws-general-policies/ensu
re-that-s3-buckets-are-encrypted-with-kms-by-default
Check: CKV2_AWS_6: "Ensure that S3 bucket has a Public Access block"
        FAILED for resource: aws_s3_bucket.this
        File: \main.tf:1-5
```

📊 Résultat attendu

Checkov va détecter :

- ❌ bucket public
- ❌ pas de chiffrement
- ❌ pas de versioning
- ❌ pas de logging

🟡 Étape 4 — Correction (DevSecOps loop)

4.1 Correction – première étape

On modifie le code avec la version sécurisée (voir Annexe 1) . On remplace main.tf par le main.tf de l'Annexe 1

Puis à nouveau :

```
terraform plan
checkov -d .    ou bien selon le cas : py -3.11 -m checkov.main -d .
```

👉 jusqu'à ce que ce soit propre

Après application des commandes : plan et checkov, on constate que le nombre de failed a beaucoup diminué, Les seuls failed qui restent sont :

```
Check: CKV2_AWS_62: "Ensure S3 buckets should have event notifications enabled"
        FAILED for resource: aws_s3_bucket.this
        File: \main.tf:1-5
        Guide: https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/aws-logging-policies/bc-aws-2-62
Check: CKV2_AWS_61: "Ensure that an S3 bucket has a lifecycle configuration"
        FAILED for resource: aws_s3_bucket.this
        File: \main.tf:1-5
        Guide: https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/aws-logging-policies/bc-aws-2-61
Check: CKV_AWS_18: "Ensure the S3 bucket has access logging enabled"
        FAILED for resource: aws_s3_bucket.this
        File: \main.tf:1-5
        Guide: https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/s3-policies/s3-13-enable-logging
Check: CKV_AWS_144: "Ensure that S3 bucket has cross-region replication enabled"
        FAILED for resource: aws_s3_bucket.this
        File: \main.tf:1-5
        Guide: https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/aws-general-policies/ensure-that-s3-bucket-has-cross-region-replication-enabled
Check: CKV_AWS_145: "Ensure that S3 buckets are encrypted with KMS by default"
        FAILED for resource: aws_s3_bucket.this
        File: \main.tf:1-5
        Guide: https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/aws-general-policies/ensure-that-s3-buckets-are-encrypted-with-kms-by-default
PS C:\terraform\terraform-checkov-lab>
PS C:\terraform\terraform-checkov-lab>
```

4.2 CKV_AWS_18

On vas corriger dans le main.tf le warning : CKV_AWS_18 — Logging .
Voir l'Annexe 2 .
On ajoute dans le main.tf le bloc de code suivant :

```
resource "aws_s3_bucket" "logs" {
  bucket = "${var.bucket_name}-logs"
}

resource "aws_s3_bucket_logging" "this" {
  bucket        = aws_s3_bucket.this.id
  target_bucket = aws_s3_bucket.logs.id
  target_prefix = "log/"
}
```

4.3 CKV2_AWS_61 — Lifecycle

On vas corriger dans le main.tf le warning : CKV2_AWS_61— Lifecycle

```
resource "aws_s3_bucket_lifecycle_configuration" "this" {
  bucket = aws_s3_bucket.this.id

  rule {
    id     = "cleanup"
    status = "Enabled"

    expiration {
      days = 30
    }
  }
}
```

4.4 autres corrections optionnelles

CKV_AWS_144 — Cross-region replication
(nécessite IAM + versioning + autre bucket)
CKV2_AWS_62 — Event notifications
(nécessite Lambda / SNS / SQS)

4.5 Enlever des exceptions

Dans `.checkov.yaml` :

```
skip-check:
  - CKV_AWS_144
  - CKV2_AWS_62
```

4.6 tests finaux :

Utiliser le code de l'annexe 2 . remplacer main.tf par ce code
Ensuite modifier le fichier checkov.yaml
Nouveau contenu de ce fichier :

```
framework:
  - terraform

output: cli

skip-check:
  - CKV_AWS_144   # cross-region replication (hors scope)
  - CKV2_AWS_62   # event notifications (hors scope)
```

Puis :

```
terraform plan
checkov -d .    ou bien selon le cas : py -3.11 -m checkov.main -d .
```

4.6.1 Résultats :

- ✅ Passed: **25**
- ❌ Failed: **8**

👉 Et surtout : 👉 **aucune fail critique majeure**

```
Check: CKV_AWS_300: "Ensure S3 lifecycle configuration sets period for aborting failed uploads"
      FAILED for resource: aws_s3_bucket_lifecycle_configuration.this
      File: \main.tf:92-103
      Guide:
https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/aws-general-policies
/bc-aws-300

            92  | resource "aws_s3_bucket_lifecycle_configuration" "this" {
            93  |   bucket = aws_s3_bucket.this.id
            94  |
            95  |   rule {
            96  |     id     = "cleanup"
            97  |     status = "Enabled"
            98  |
            99  |     expiration {
            100 |       days = 30
            101 |     }
            102 |   }
```

```
        103 | }

Check: CKV_AWS_300: "Ensure S3 lifecycle configuration sets period for aborting failed uploads"
    FAILED for resource: aws_s3_bucket_lifecycle_configuration.logs
    File: \main.tf:105-116
    Guide:
https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/aws-general-policies
/bc-aws-300

        105 | resource "aws_s3_bucket_lifecycle_configuration" "logs" {
        106 |   bucket = aws_s3_bucket.logs.id
        107 |
        108 |   rule {
        109 |     id     = "cleanup"
        110 |     status = "Enabled"
        111 |
        112 |     expiration {
        113 |       days = 30
        114 |     }
        115 |   }
        116 | }
Check: CKV2_AWS_62: "Ensure S3 buckets should have event notifications enabled"
    FAILED for resource: aws_s3_bucket.this
    File: \main.tf:4-7
    Guide:
https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/aws-logging-policies/
bc-aws-2-62

        4 | resource "aws_s3_bucket" "this" {
        5 |   bucket = var.bucket_name
        6 |   tags   = var.tags
        7 | }

Check: CKV2_AWS_62: "Ensure S3 buckets should have event notifications enabled"
    FAILED for resource: aws_s3_bucket.logs
    File: \main.tf:12-15
    Guide:
https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/aws-logging-policies/
bc-aws-2-62

        12 | resource "aws_s3_bucket" "logs" {
        13 |   bucket = "${var.bucket_name}-logs"
        14 |   tags   = var.tags
        15 | }

Check: CKV_AWS_144: "Ensure that S3 bucket has cross-region replication enabled"
    FAILED for resource: aws_s3_bucket.this
    File: \main.tf:4-7
```

```
    Guide:
https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/aws-general-policies
/ensure-that-s3-bucket-has-cross-region-replication-enabled

        4 | resource "aws_s3_bucket" "this" {
        5 |   bucket = var.bucket_name
        6 |   tags   = var.tags
        7 | }

Check: CKV_AWS_144: "Ensure that S3 bucket has cross-region replication enabled"
    FAILED for resource: aws_s3_bucket.logs
    File: \main.tf:12-15
    Guide:
https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/aws-general-policies
/ensure-that-s3-bucket-has-cross-region-replication-enabled

        12 | resource "aws_s3_bucket" "logs" {
        13 |   bucket = "${var.bucket_name}-logs"
        14 |   tags   = var.tags
        15 | }

Check: CKV_AWS_145: "Ensure that S3 buckets are encrypted with KMS by default"
    FAILED for resource: aws_s3_bucket.this
    File: \main.tf:4-7
    Guide:
https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/aws-general-policies
/ensure-that-s3-buckets-are-encrypted-with-kms-by-default

        4 | resource "aws_s3_bucket" "this" {
        5 |   bucket = var.bucket_name
        6 |   tags   = var.tags
        7 | }

Check: CKV_AWS_145: "Ensure that S3 buckets are encrypted with KMS by default"
    FAILED for resource: aws_s3_bucket.logs
    File: \main.tf:12-15
    Guide:
https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/aws-general-policies
/ensure-that-s3-buckets-are-encrypted-with-kms-by-default

        12 | resource "aws_s3_bucket" "logs" {
        13 |   bucket = "${var.bucket_name}-logs"
        14 |   tags   = var.tags
        15 | }
```

-

🟡 Étape 5 — Apply

```
terraform apply -var-file=environments/dev/terraform.tfvars
```

Annexe 1 – version sécurisée

👉 Remplace `main.tf` par :

```
resource "aws_s3_bucket" "this" {
  bucket = var.bucket_name

  tags = var.tags
}

# ✅ Bloquer accès public
resource "aws_s3_bucket_public_access_block" "this" {
  bucket = aws_s3_bucket.this.id

  block_public_acls       = true
  block_public_policy     = true
  ignore_public_acls      = true
  restrict_public_buckets = true
}

# ✅ Versioning
resource "aws_s3_bucket_versioning" "this" {
  bucket = aws_s3_bucket.this.id

  versioning_configuration {
    status = "Enabled"
  }
}

# ✅ Encryption
resource "aws_s3_bucket_server_side_encryption_configuration" "this" {
  bucket = aws_s3_bucket.this.id

  rule {
    apply_server_side_encryption_by_default {
      sse_algorithm = "AES256"
    }
  }
}
```

Annexe 2 – Version finale

Code final propre :

```
############################
# MAIN BUCKET
############################
resource "aws_s3_bucket" "this" {
  bucket = var.bucket_name
  tags   = var.tags
}

############################
# LOGS BUCKET
############################
resource "aws_s3_bucket" "logs" {
  bucket = "${var.bucket_name}-logs"
  tags   = var.tags
}

############################
# PUBLIC ACCESS BLOCK
############################
resource "aws_s3_bucket_public_access_block" "this" {
  bucket = aws_s3_bucket.this.id

  block_public_acls       = true
  block_public_policy     = true
  ignore_public_acls      = true
  restrict_public_buckets = true
}

resource "aws_s3_bucket_public_access_block" "logs" {
  bucket = aws_s3_bucket.logs.id

  block_public_acls       = true
  block_public_policy     = true
  ignore_public_acls      = true
  restrict_public_buckets = true
}

############################
# VERSIONING
```

```
############################
resource "aws_s3_bucket_versioning" "this" {
  bucket = aws_s3_bucket.this.id

  versioning_configuration {
    status = "Enabled"
  }
}

resource "aws_s3_bucket_versioning" "logs" {
  bucket = aws_s3_bucket.logs.id

  versioning_configuration {
    status = "Enabled"
  }
}

############################
# ENCRYPTION (AES256)
############################
resource "aws_s3_bucket_server_side_encryption_configuration" "this" {
  bucket = aws_s3_bucket.this.id

  rule {
    apply_server_side_encryption_by_default {
      sse_algorithm = "AES256"
    }
  }
}

resource "aws_s3_bucket_server_side_encryption_configuration" "logs" {
  bucket = aws_s3_bucket.logs.id

  rule {
    apply_server_side_encryption_by_default {
      sse_algorithm = "AES256"
    }
  }
}

############################
# ACCESS LOGGING
############################
resource "aws_s3_bucket_logging" "this" {
  bucket        = aws_s3_bucket.this.id
  target_bucket = aws_s3_bucket.logs.id
  target_prefix = "log/"
}
```

```
############################
# LIFECYCLE POLICY
############################
resource "aws_s3_bucket_lifecycle_configuration" "this" {
  bucket = aws_s3_bucket.this.id

  rule {
    id     = "cleanup"
    status = "Enabled"

    expiration {
      days = 30
    }
  }
}

resource "aws_s3_bucket_lifecycle_configuration" "logs" {
  bucket = aws_s3_bucket.logs.id

  rule {
    id     = "cleanup"
    status = "Enabled"

    expiration {
      days = 30
    }
  }
}
```